



Cytology

Dr. Reshna K. Albarzanji

B.Sc. M.Sc. PhD.

Cytogenetics

Hawler Medical University

College of Medicine

Microbiology dep.

reshna.kamal@hmu.edu.krd

SGL : 5

*Structure ,Position and function of
cell organelles*

Peroxisome

"Microbody"

small ($d = 0.1 - 1 \mu\text{m}$)

Outer lipid bilayer, inside finely granulated substance

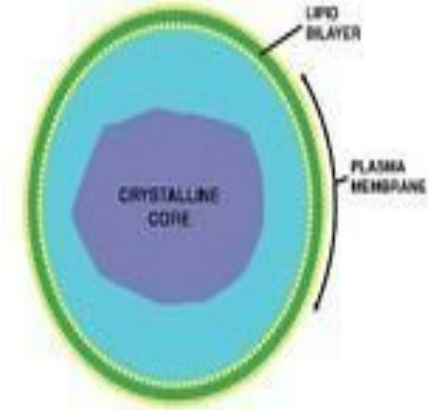
in some mammals (e.g., rat), the core crystallized in the middle (consisting of urate oxidase)

Contains 2 enzymes: peroxidase and catalase

Occurrence: all cells, but especially in those where detoxification (liver cells, renal tubules) or intensive lipid metabolism (myelin sheath, sebaceous gland).

Origin: either from ER or by division from an existing peroxisome

Proteins are produced in the cytosol on free ribosomes and with the help of peroxins (different proteins) are transported the organelle.



Anatomy of the Peroxisome

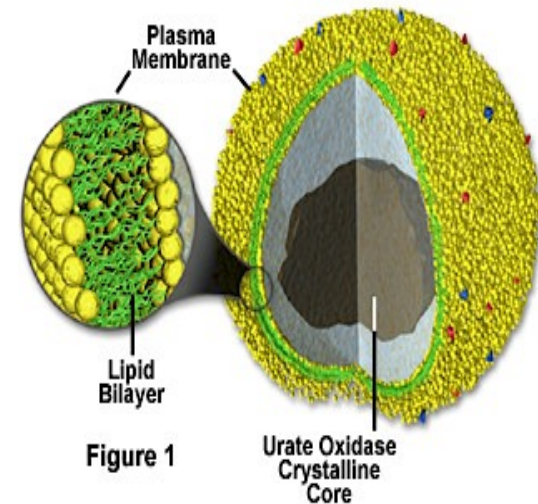
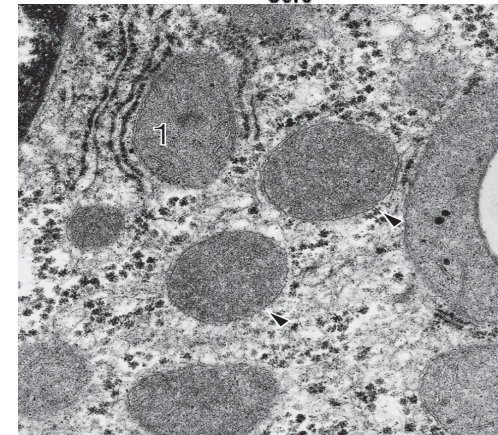
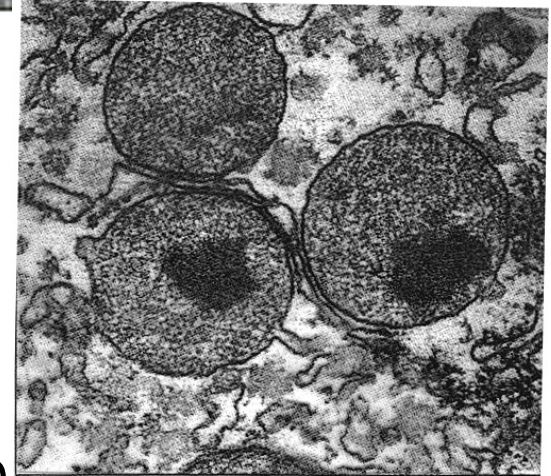
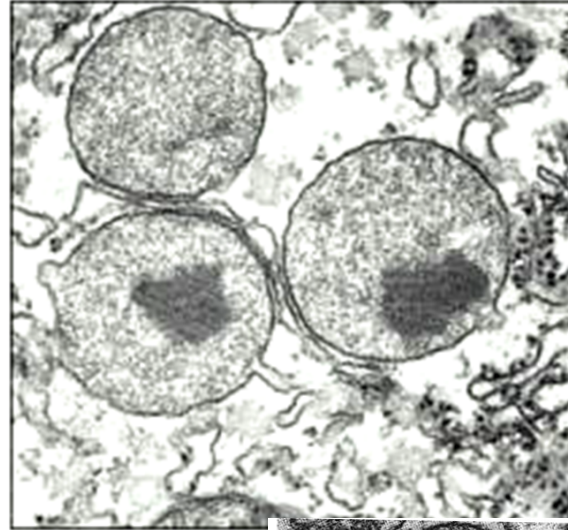


Figure 1



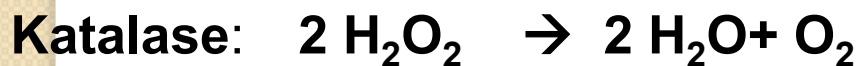
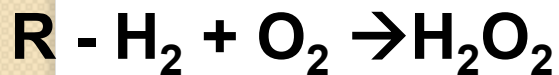
Function

Lipid synthesis and oxidation
Detoxification (liver, kidney):
phenol, ethanol, formaldehyde
Protection against oxygen radicals



Enzymes

Peroxidase: withdraw H from molecules using O_2



Other enzymes:

Urine oxidase, fatty acid oxidases, D-amino acid oxidase

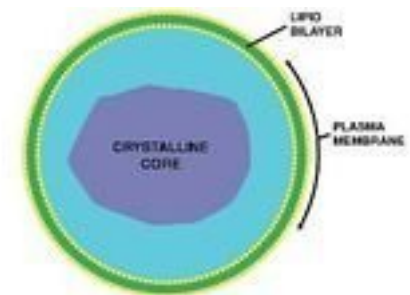
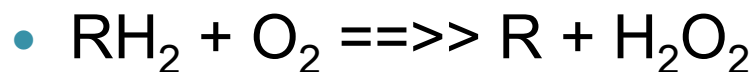
- Peroxisomes
- Functions

1- Peroxisomes are involved in the formation and decomposition of hydrogen peroxide and the word peroxisome is actually derived from hydrogen peroxide.

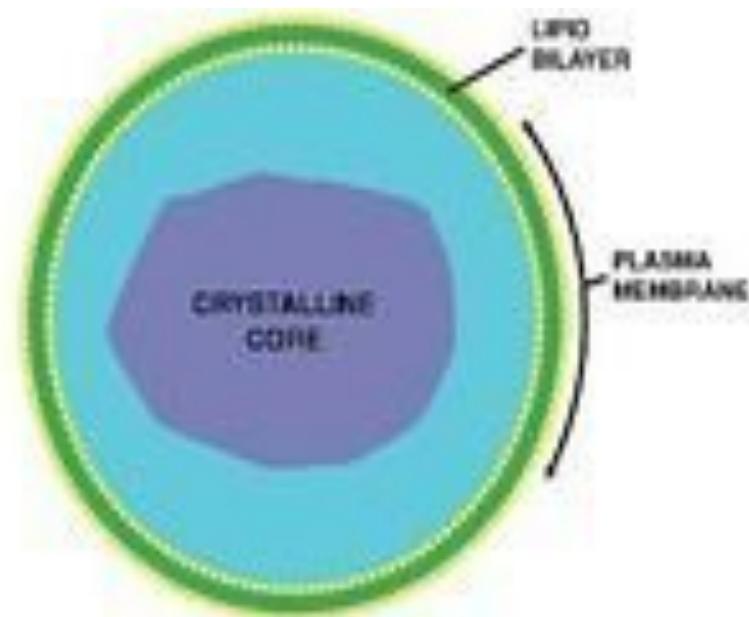
Peroxisome contains oxidative enzymes, such as catalase, D-amino acid oxidase and uric acid oxidase.

They use molecular oxygen to remove hydrogen atoms from a specific organic substrate (R) in an oxidative reaction.

It produces hydrogen peroxide (H_2O_2 is a toxic byproduct of cellular metabolism).



- Catalases uses this H_2O_2 in the peroxisome to oxidize other substrates like phenols, formic acid, formaldehyde, and alcohol: $\text{H}_2\text{O}_2 + \text{R}'\text{H}_2 \Rightarrow \text{R}' + 2\text{H}_2\text{O}$
- This reaction is important in liver and kidney cells where the peroxisomes detoxify various toxic substances that enter the blood.
- In addition, when excess H_2O_2 accumulates in the cell, catalase converts it to H_2O through this reaction: $2\text{H}_2\text{O}_2 \Rightarrow 2\text{H}_2\text{O} + \text{O}_2$





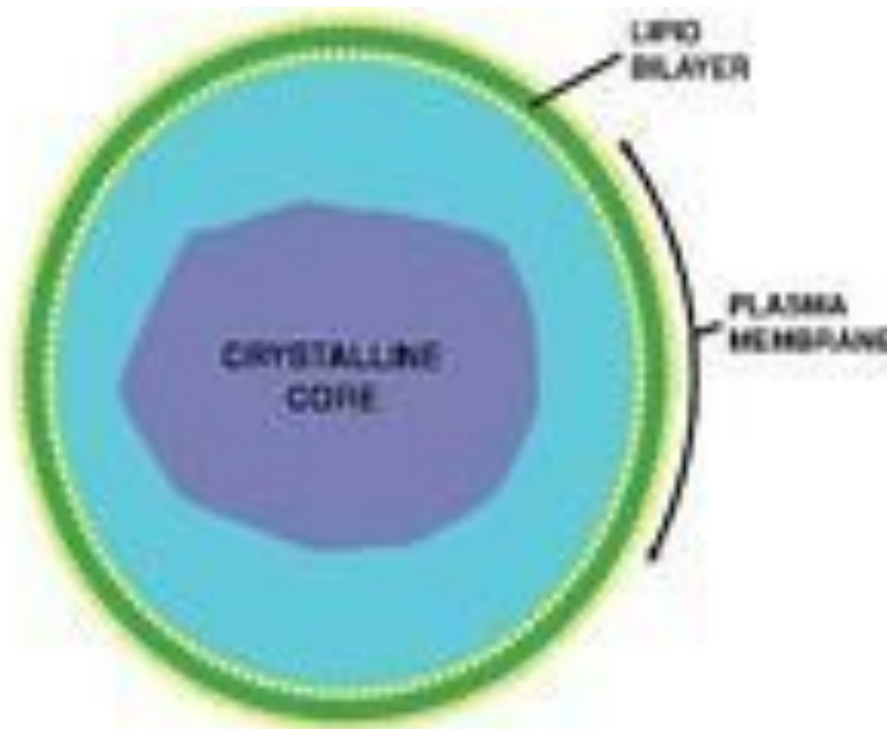
2- Peroxisomes are important for lipids metabolism. In humans, oxidation of fatty acids greater than 18 carbons in length occurs in peroxisomes.

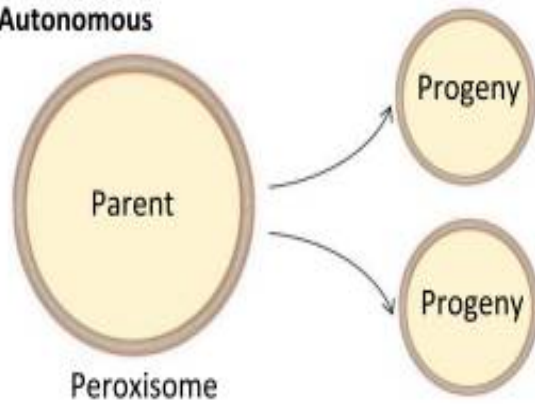
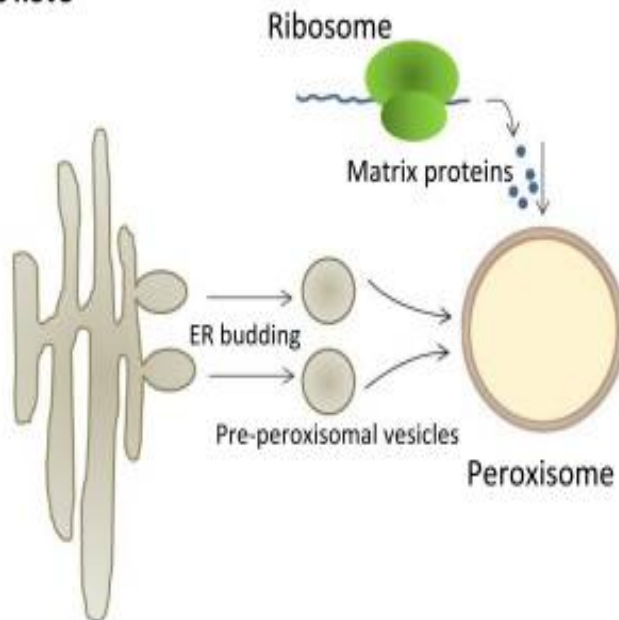
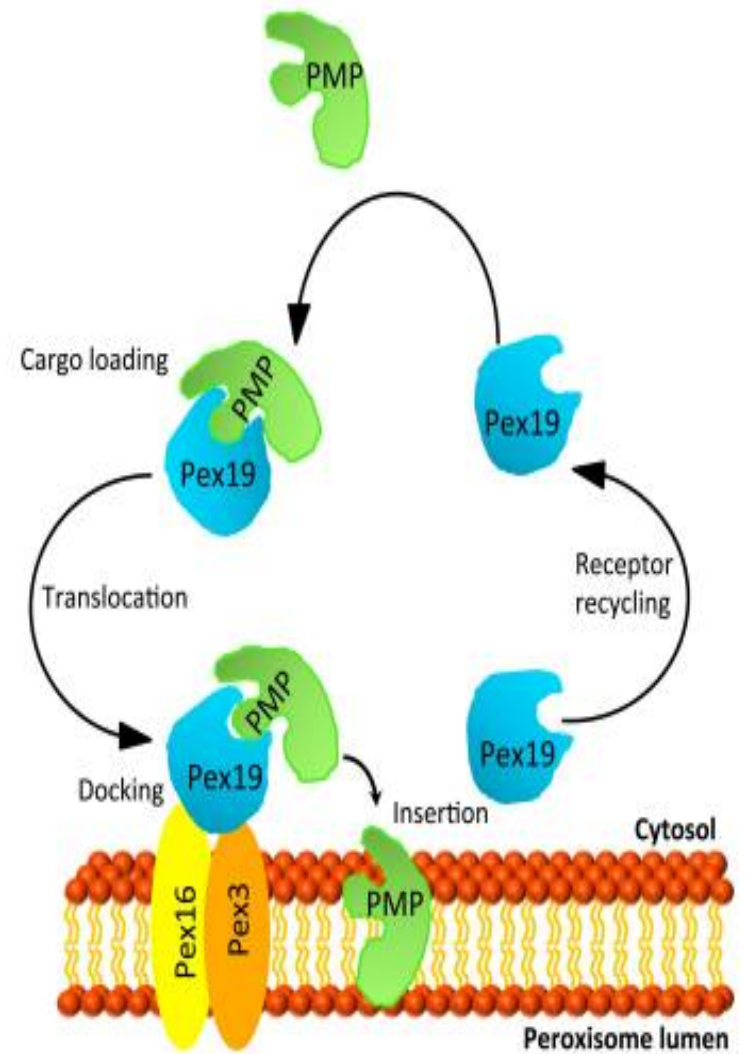
A major function of peroxisomes is the breakdown of fatty acid molecules in a process called beta-oxidation. In this process, the fatty acids are broken down into Acetyl-CoA.

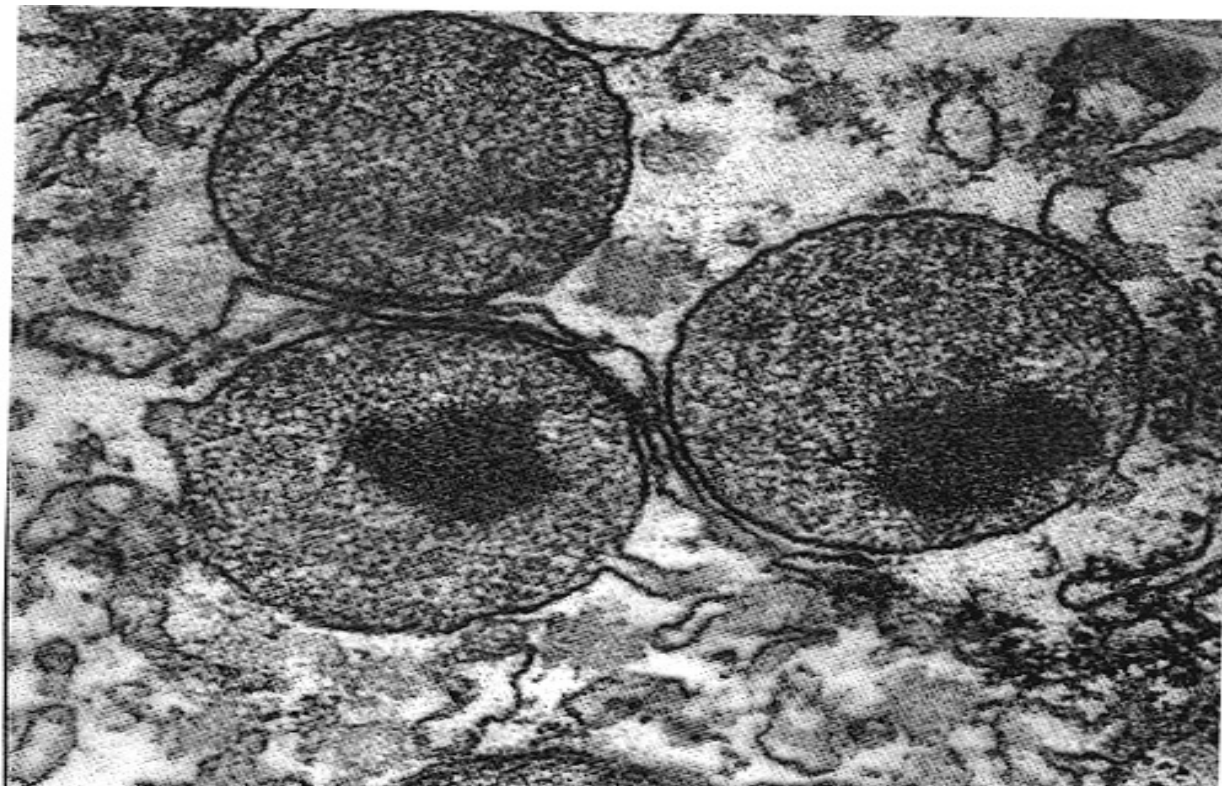
It is then transported to back to the cytosol for further use.

3- Peroxisomes contain the first two enzymes required for the synthesis of plasmalogens (myelin sheath).

4- Peroxisomes also play important roles in cholesterol and bile acid synthesis, and prostaglandin metabolism.

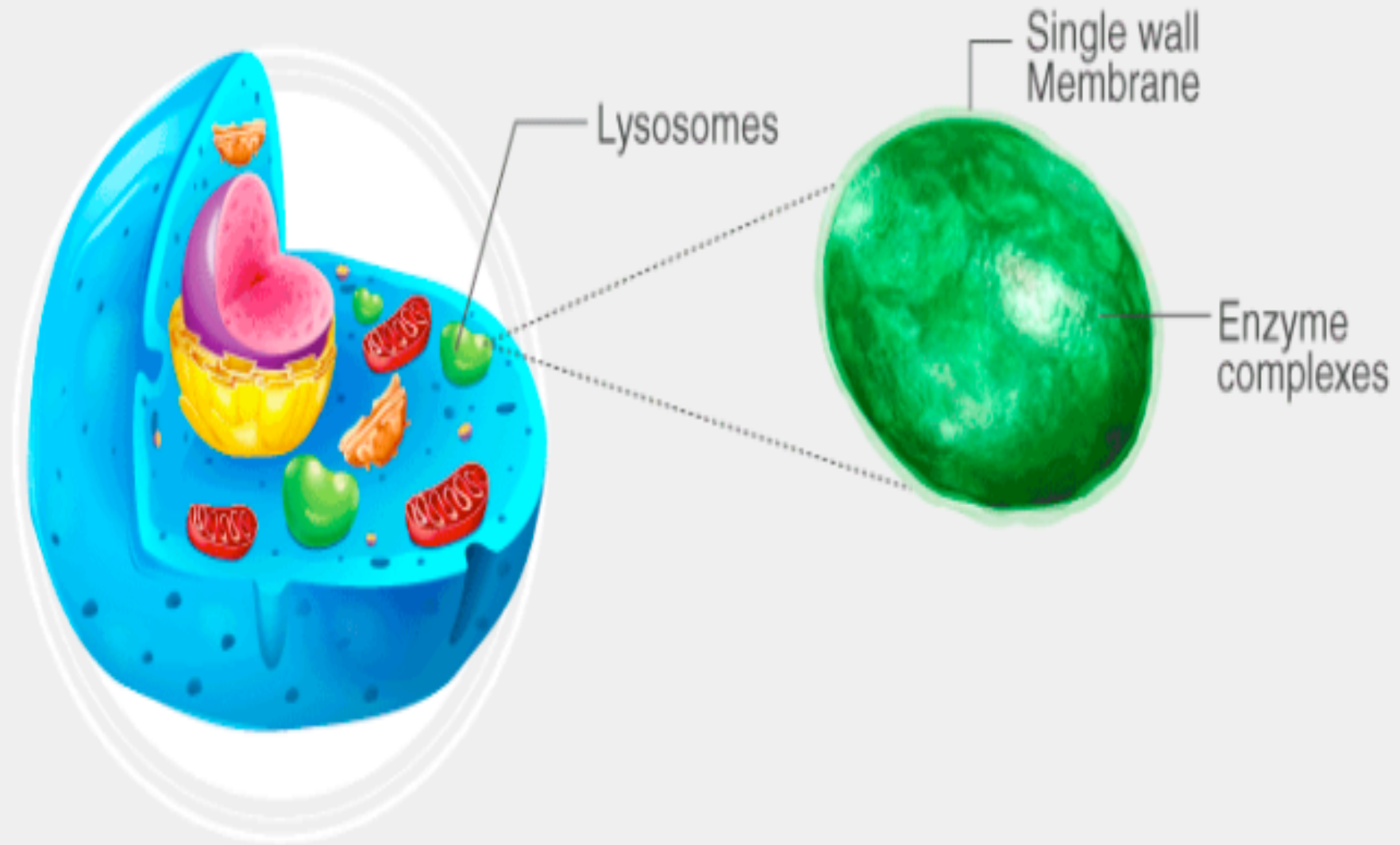


A**Autonomous****De novo****B**



LYSOSOMES

- Lysosomes are an important cell organelle found within eukaryotic animal cells. they are also known as the “*suicide bags*” of the cell.
- The term was coined by Christian de Duve, a Belgian biologist, who discovered it and ultimately got a Nobel Prize in Medicine or Physiology in the year 1974.
- ***Lysosomes are sphere-shaped sacs filled with hydrolytic enzymes that have the capability to break down many types of biomolecules.”***



- **Lysosome Structure**

- Lysosomes are membrane-bound organelles and the area within the membrane is called the lumen, which contains the hydrolytic enzymes and other cellular debris.
- The pH level of the lumen lies between 4.5 and 5.0, which makes it quite acidic. It is almost comparable to the function of acids found in the stomach.
- The sizes of lysosomes vary, with the largest ones measuring in more at than 1.2 μm . But they typically range from 0.1 μm to 0.6 μm .

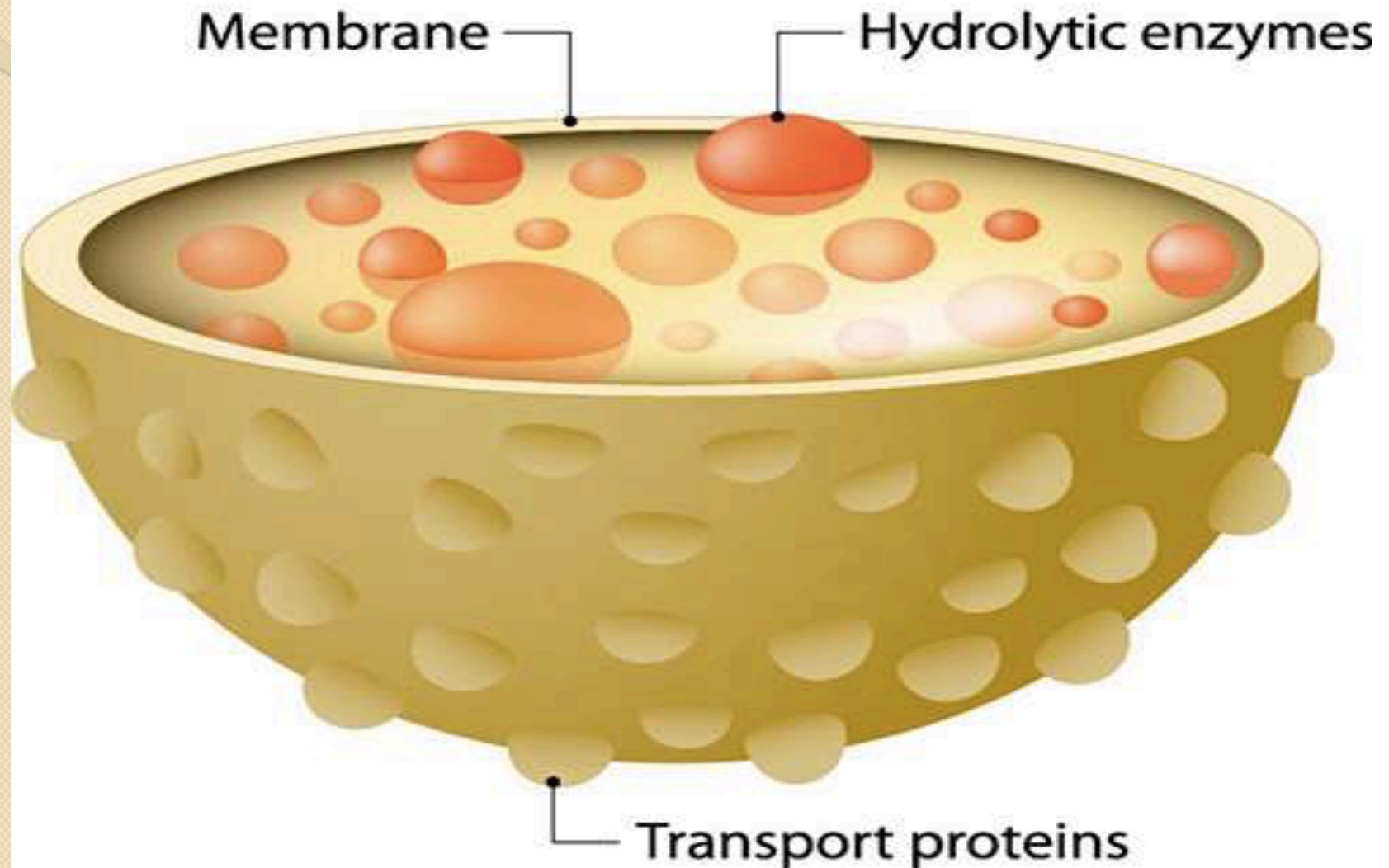
Functions of lysosomes

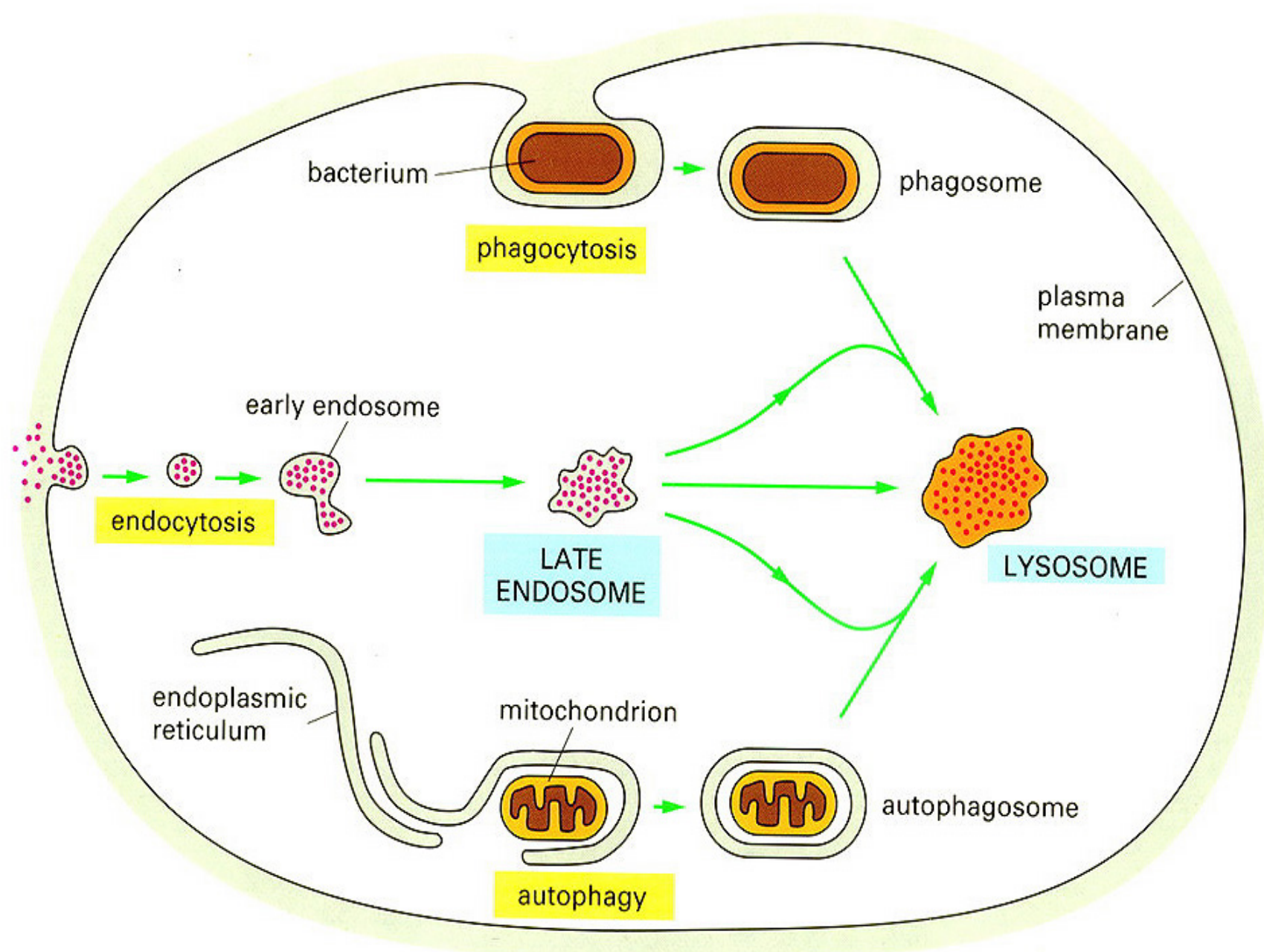
I. Intracellular Digestion

- Within the secondary lysosomes, the ingested material or those resulting from autophagy are subjected to reaction of many hydrolases. Carbohydrates are hydrolysed to monosaccharides, proteins into dipeptides
- **2. Autophagy**
- Autophagy is the natural, conserved degradation of the cell that removes unnecessary or dysfunctional components through a lysosome-dependent regulated mechanism. It allows the orderly degradation and recycling of cellular components

- **3. Removal of cell and extra cellular material during developmental process**
- Lysosome bring about shedding or remodeling of tissues with the removal of whole cells and extra cellular material during developmental process
- **4. Lysosomes important in germ cells and fertilization**
- The acrosome of spermatozoa can be considered as the specialized lysosome which contains PROTEASE and HYALURONIDASE and abundant ACID PHOSPHATE. During fertilization, hyaluronidase and protease digests zone pellucide making a channel through which the sperm nucleus penetrates the egg.

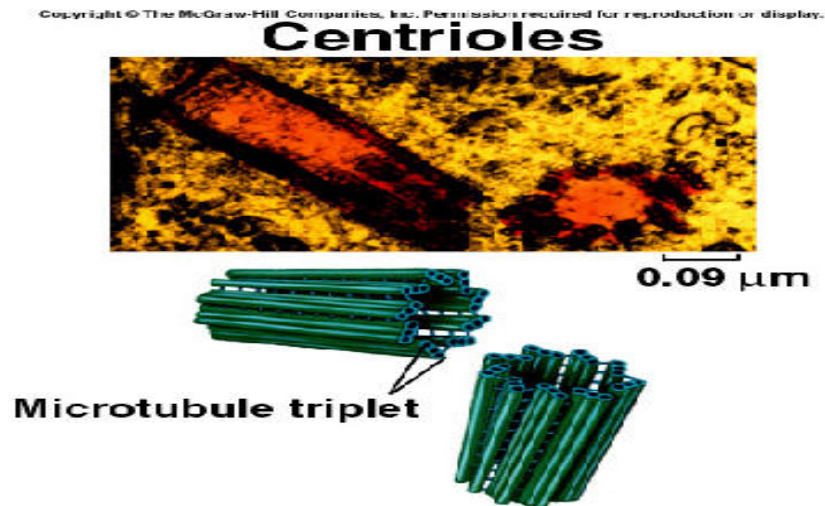
LYSOSOME





Centrosomes and Centrioles

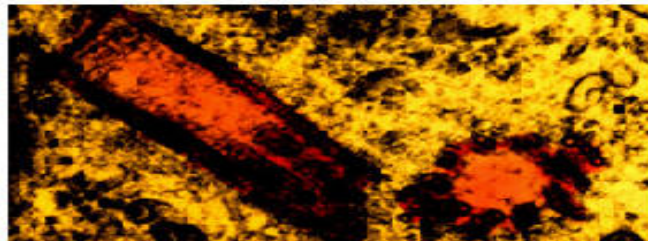
- Are paired, short rod like cytoplasmic bodies, early cytologists recognized centrioles as forming the ends of the mitotic spindle, in non dividing cells its found close to the nucleus.
- Centrioles anchor spindle fibers during cell division



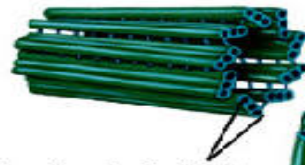
- The electron microscope reveals that each rod shaped centriole is about 0.2 μm long & consist of nine triples of microtubules that are oriented parallel to the long axis of the organelle

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Centrioles



0.09 μm



Microtubule triplet



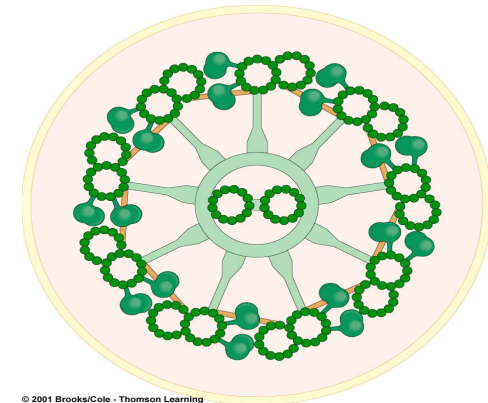
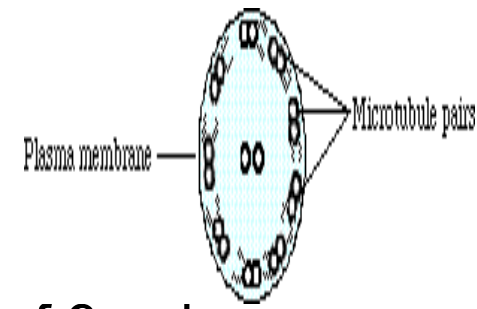
Basal bodies:

- Each cilium must have a basal body,
- Basal bodies are barrel-like microtubular structures located near the cell surface.
- The structure of basal bodies is **similar to that of centrioles**
- basal bodies are produced by the repeated replication of centrioles and the migration of the newly formed organelles to the apical surface of the cell, each centriole derived basal body, serves as the organizer for the microtubules of a cilium.

Cilia and Flagella

Structure

- Each cilium (or flagellum) is a circular series of 9 pairs of microtubules (often containing an additional central pair of tubules) surrounded by a long slender sheath of plasma membrane.
- This structure is called an axoneme. The nine outer pairs are made up of motor proteins called dynein. These are large and flexible that allows the cilia to move.
- Cilia are attached to the cell at the basal body that is made up of microtubules arranged in nine triplets.
- They are very minute structures ranging from $0.25\ \mu\text{m}$ in diameter to $20\ \mu\text{m}$ in length



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Flagella and Cilia



Flagellum



Basal body

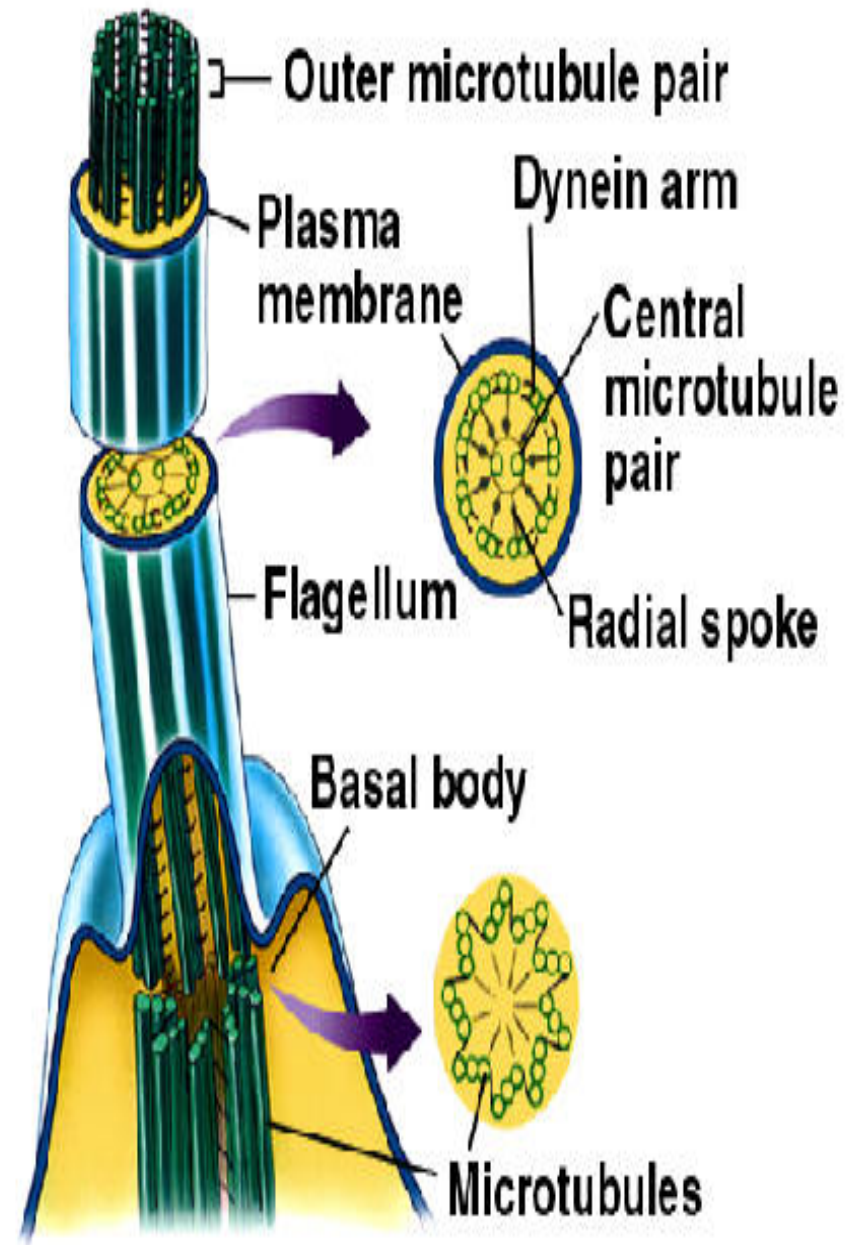
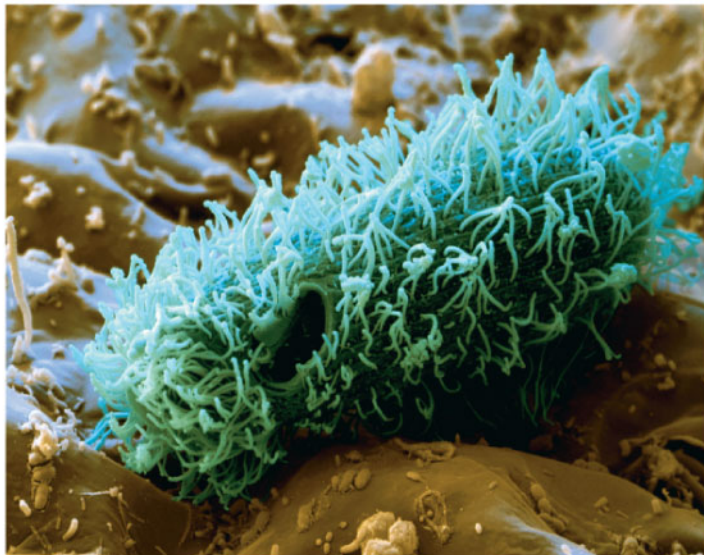


Paramecium 4.36 μm

Function

- Cilia and flagella are used primarily by animal cells for movement.
- Flagella are larger single versions of cilia.
- Cilia are short but numerous compared to flagella.

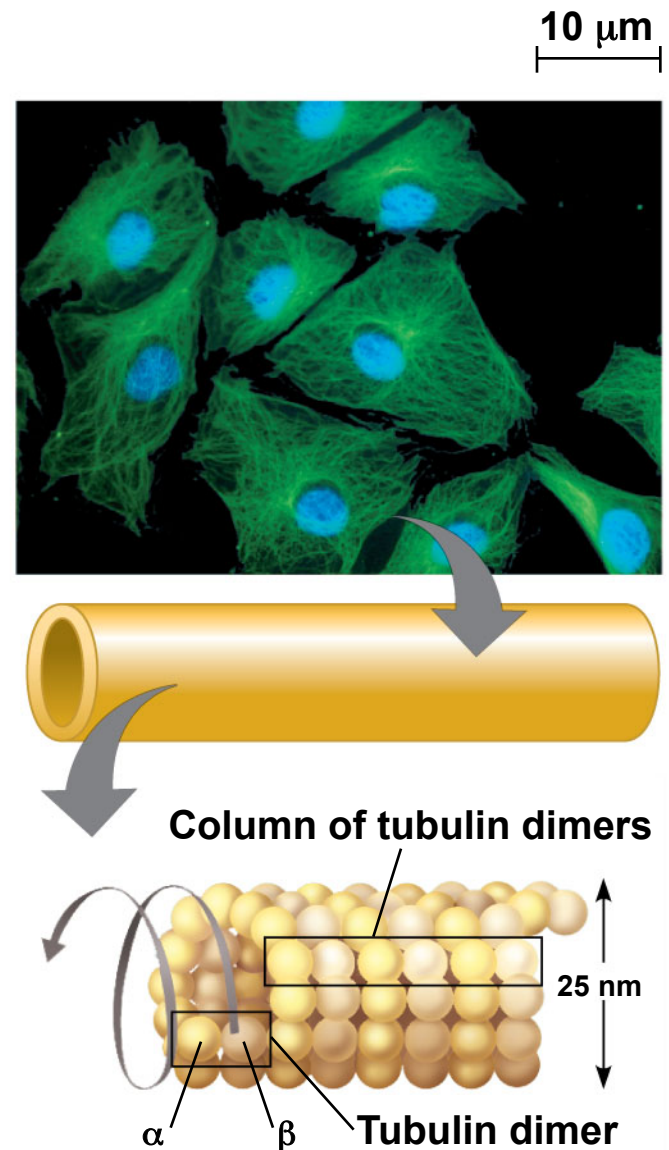
Eukaryotic Flagellum

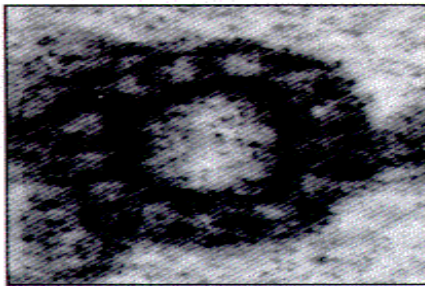


- **Cytoskeleton:**

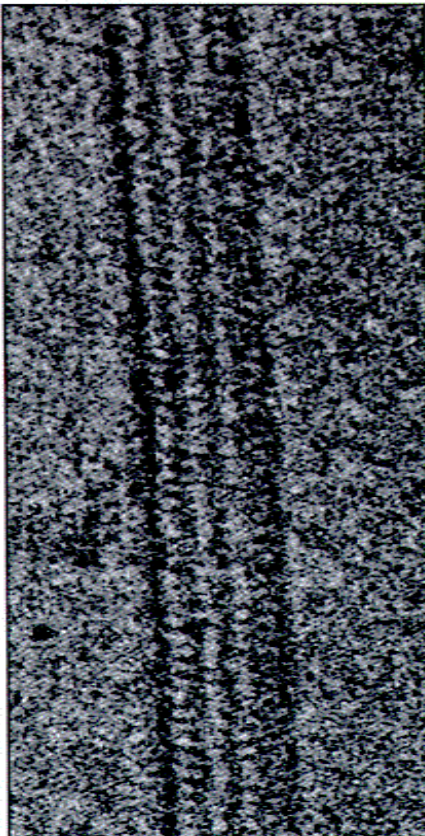
- A system of filaments or fibers that is present in the cytoplasm of eukaryotic cells (cells containing a nucleus). The cytoskeleton organizes other constituents of the cell, maintains the cell's shape, and is responsible for the locomotion of the cell itself and the movement of the various organelles within it.
- Its composed of three parts
- 1- microtubules
- 2- microfilaments
- 3- intermediate filaments

Property	Microtubules (Tubulin Polymers)
Structure	Hollow tubes; wall consists of 13 columns of tubulin molecules
Diameter	25 nm with 15-nm lumen
Protein subunits	Tubulin, a dimer consisting of α -tubulin and β -tubulin
Main functions	Maintenance of cell shape (compression-resisting “girders”) Cell motility (as in cilia or flagella) Chromosome movements in cell division Organelle movements

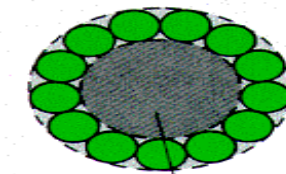




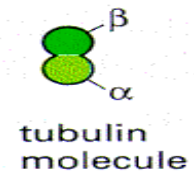
(A)



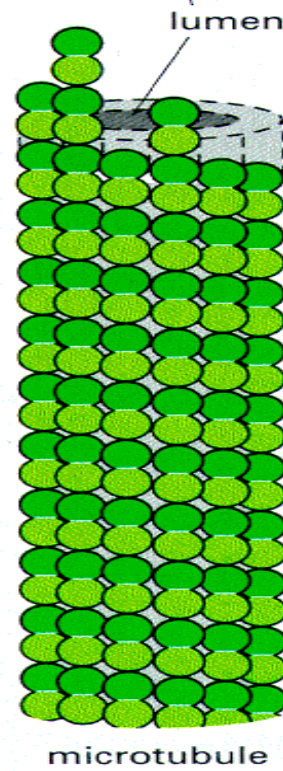
(B)



(C)



tubulin molecule



(D)

protofilament



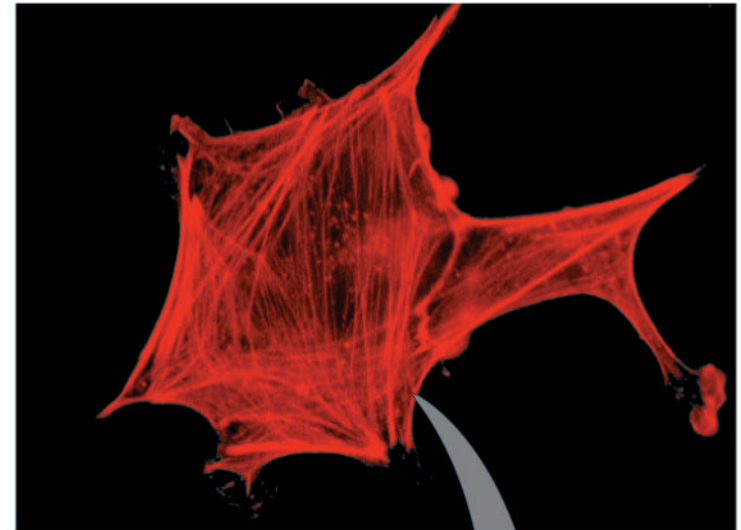
(E)

plus end

minus end

Property	Microfilaments (Actin Filaments)
Structure	Two intertwined strands of actin, each a polymer of actin subunits
Diameter	7 nm
Protein subunits	Actin
Main functions	Maintenance of cell shape (tension-bearing elements) Changes in cell shape Muscle contraction Cytoplasmic streaming Cell motility (as in pseudopodia) Cell division (cleavage furrow formation)

10 μm

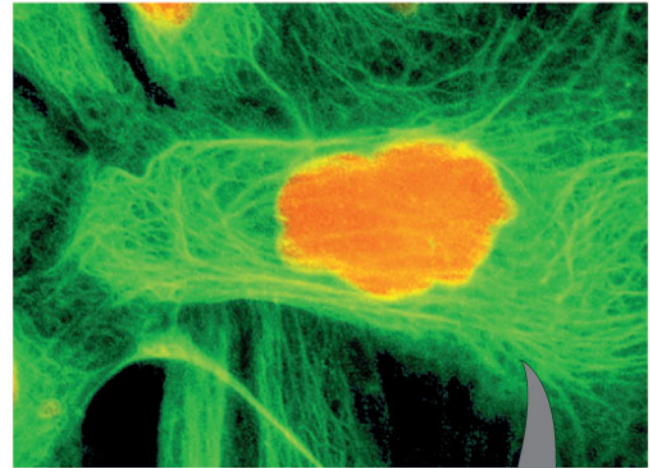


Actin subunit



Property	Intermediate Filaments
Structure	Fibrous proteins supercoiled into thicker cables
Diameter	8–12 nm
Protein subunits	One of several different proteins (such as keratins), depending on cell type
Main functions	Maintenance of cell shape (tension-bearing elements) Anchorage of nucleus and certain other organelles Formation of nuclear lamina

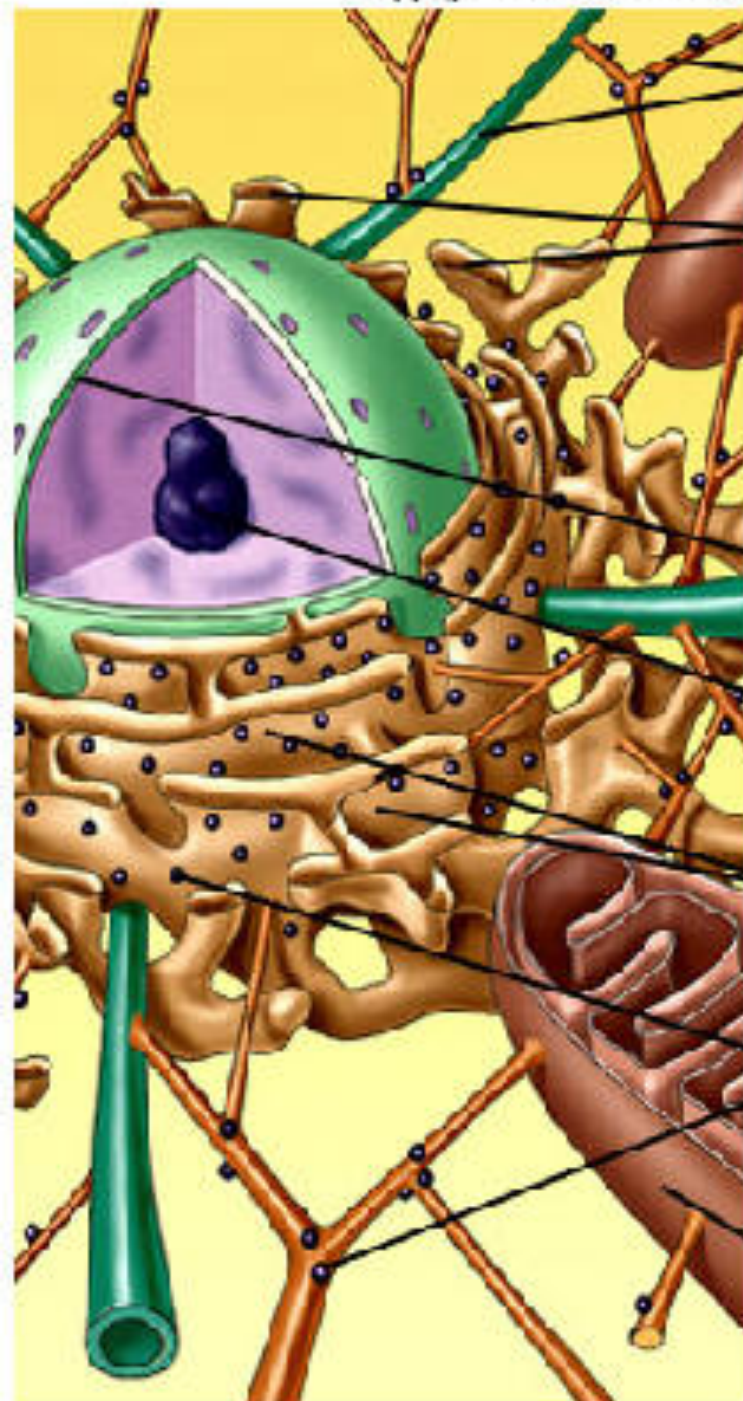
5 μm



Keratin proteins

Fibrous subunit (keratins coiled together)





Cytoskeleton

**Smooth
endoplasmic
reticulum**

Cytoskeleton

**Nuclear
envelope**

Nucleolus

**Rough
endoplasmic
reticulum**

Ribosomes

Mitochondrion

12-Vacuole

Structure

- Vacuoles are membrane bound sacs with little or no internal structure.
- Vacuoles are generally used to store cell products.
- The large central vacuole of plant cells is a special case. It is used to store water and cellular wastes and helps maintain the necessary internal pressure

Inclusions

- contain accumulated metabolites or other substances, but unlike organelles have little or no metabolic activity themselves. Most inclusions are transitory structures not enclosed by membrane. Features of important and commonly seen inclusions are shown in Figure 2–28 and include the following:
- **Lipid droplets**, accumulations of lipid-filling adipocytes

13-Inclusions:

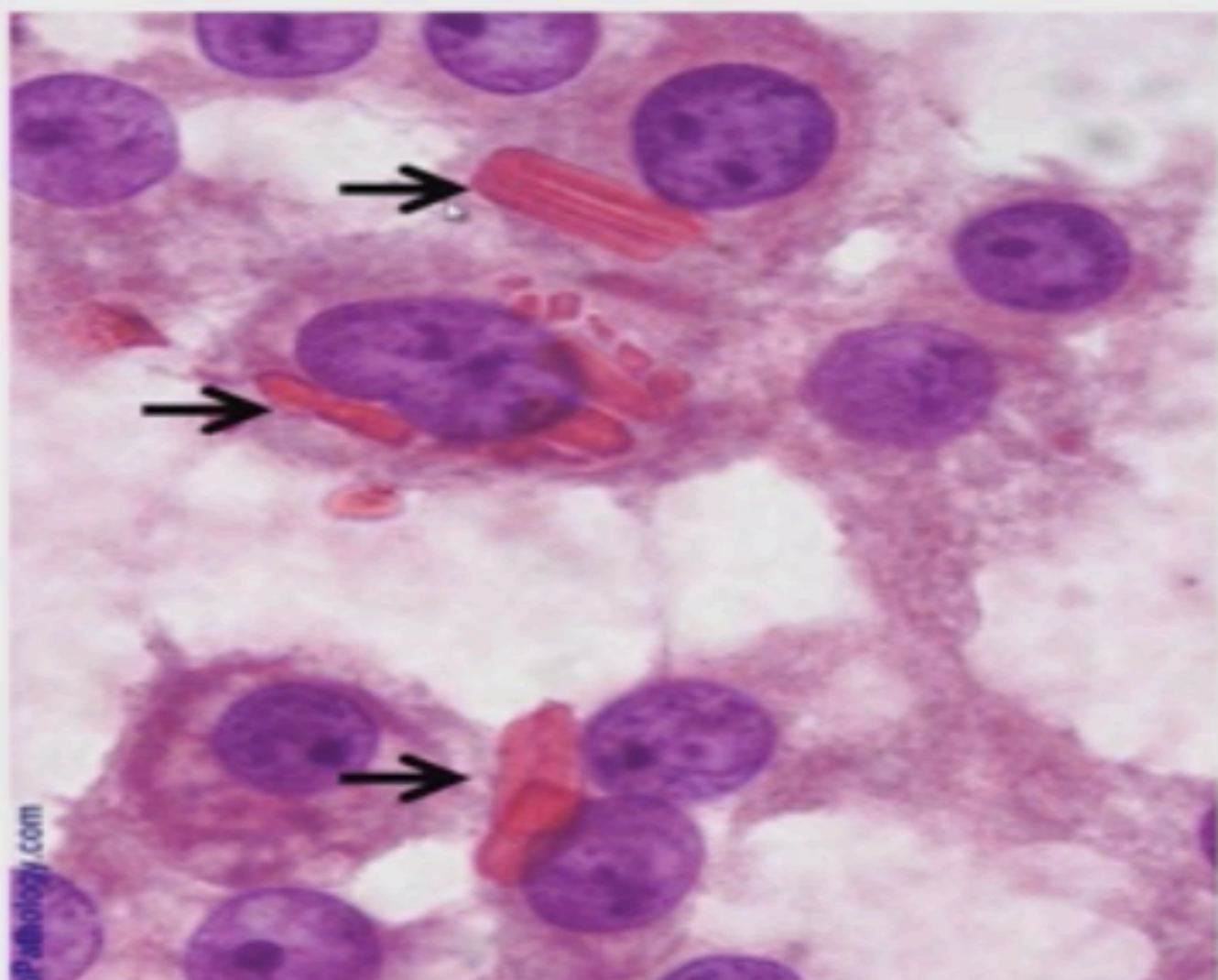
- Are considered to be non living components of the cell that do not possess metabolic activity and are not bounded by membrane .The most common inclusions are Glycogen ,Lipid droplets, pigments and crystals.
- **a-Glycogen:**
- **Glycogen granules**, aggregates of the carbohydrate polymer in which glucose is stored, visible as irregular clumps in muscle and liver cells.

13-Inclusions:

- **b- Lipids:**
 - Are storage form of triglycerides, not only stored in specialized cells (Adipocytes) but also are located as individual droplets in various cell types
- **c-Pigments:**
 - The most common pigment in the body beside hemoglobin of red blood cells is melanin manufactured by melanocytes of skin and hair, pigment cells of the retina, these pigments have protective functions in skin.

13-Inclusions:

- 4-Crystals:
- Are not commonly found in the cells except in sertoli cells (crystals of Charcot Böttcher), interstitial cells (Crystal of Reinke) of the testes.





- * What is peroxisome
- * why this organelle named as peroxisome ?
- * which enzymes are found in peroxisomes?
- * what is the important of each enzyme?
- * how peroxisomes produced inside the cell?
- *
- * What is a lysosome.
- * Enumerate important functions of lysosome.
- * What is the main constituent of centrioles?
- * What is the function of centrioles in dividing cells?
- * What is basal body and why they are found in some cells?

- *What is the main structure of both cilia and flagella?
- *Where cilia and flagella found ?
- *Are they surrounded by plasma membrane?
- *What is the function of both of them?
- *What is cytoskeletal system?
- *In which cell this system is found?
- *Why its not found in plant cells?
- * what are the main constituents of cytoskeletal system?
- *What is microtubules made of?
- What are the main functions of microtubules?

- *What are microfilaments Structure and functions?
- *What is intermediate filament why they named like this, what function they have in cells?
- * Are vacuoles found in all animal cells? What is the structure of it?
- * What are inclusions , why they named like this?
- Enumerate different types of inclusions in animal cell?